PROJECT REPORT

GLOBAL FUEL PRICES AND TAXATION TRENDS (2020–2026)

A Tableau-Based Comparative Data Analysis

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1. Introduction

This project analyses global fuel prices from 2020 to 2026 using Tableau. It focuses on petrol, diesel, and LPG prices in different countries. It also examines Brent crude oil prices, fuel taxes, government subsidies, income levels, and regional differences. The main aim is to present this information through charts, maps, dashboards, and stories.

2. Problem Statement

Fuel prices differ across countries because of factors such as crude oil prices, taxes, government subsidies, and economic conditions. Understanding these differences can be difficult when data is presented only in tables. This project uses Tableau to make the data easier to understand through visualisations and comparisons.

3. Data Collection

The project uses a dataset named Global Fuel Prices Dataset. It includes the following information:

- Date and country
- Global region and income level
- Subsidy level
- Petrol price in USD per litre
- Diesel price in USD per litre
- LPG price in USD per litre
- Brent crude oil price
- Tax percentage

The dataset was loaded into Tableau for analysis. Source link: Global Fuel Prices Dataset (Google Drive).

4. Data Cleaning and Preparation

Before creating charts, the dataset was checked for missing values, duplicate records, incorrect values, and data type issues. Date fields and numerical fields were reviewed to ensure that they could be used correctly in Tableau. Country, region, income level, and subsidy level were treated as categories.

Note: Only the cleaning steps actually performed should be reported as completed.

5. Data Visualization

Several visualisations were created in Tableau:

- **Line charts:** Show changes in average petrol, diesel, LPG, and Brent crude prices over time.
- **World maps:** Compare petrol and diesel prices across countries.
- **Pie chart:** Shows average tax percentages across global regions.
- **Bar charts:** Compare fuel prices among selected countries and display tax percentages for selected countries.

6. Dashboard

The main Tableau dashboard combines important information in one place. It includes summary cards, line charts, a regional tax chart, a comparison of selected countries, a highest-tax-paying countries chart, and world maps for petrol and diesel prices.

AVG DIESEL PRICE	AVG PETROL PRICE	AVG FUEL TAX	AVG LPG PRICE
$2.134/L	$2.282/L	26.77%	$1.711/L

These are the values displayed in the provided dashboard screenshot and may change when filters or data selections change.



Interactive Tableau Dashboard View

Global Fuel Prices and Taxation Trends (2020–2026)

7. Tableau Story

The project contains three story perspectives designed to explain the analysis from different viewpoints.

Story 1: Dr. Sarah Mitchell – Energy Policy Consultant

This story compares petrol and diesel prices by country, average tax percentages by region, and fuel prices across subsidy levels. It helps explore differences in fuel pricing, taxation, and subsidies.

Story 2: Marcus Thompson – Global Supply Chain Manager

This story focuses on changes in average petrol, diesel, and Brent crude prices from 2020 to 2026. It also compares fuel prices among selected countries for supply-chain planning.

Story 3: Professor Elena Rodriguez – Energy Economics Researcher

This story examines tax percentages across income levels and subsidy categories. It also compares fuel prices across subsidy levels to explore relationships between taxation, income, subsidies, and fuel prices.

8. Results and Key Findings

The displayed annual average values show the following changes:

Measure	2020	2023	2026
Petrol price (USD/litre)	1.485	2.234	2.647
Diesel price (USD/litre)	1.389	2.089	2.474
Brent crude price (as displayed)	77.59	104.43	123.75
LPG price (as displayed)	1.1136	1.5754	1.9850

The charts show that the displayed average petrol and diesel prices are higher in 2026 than in 2020. Brent crude prices also rise overall across the displayed years, although some decreases occur in between. The displayed LPG values rise overall, with a decrease in 2023.

In Story 3, the average tax percentage shown for the High income-level category is 34.72, compared with 22.86 for Low and 21.11 for Middle. In the subsidy-level chart, Low displays 37.55, while High, Medium, and Very High display values around 15.

Note: These values should be checked against final Tableau worksheets before submission. Charts show patterns but do not alone prove causality.

9. Conclusion

This project demonstrates how Tableau can be used to analyse and present global fuel price data. It helps users compare prices across countries, study changes over time, and explore taxation and subsidy categories.

Through this project, I learned how to prepare data, create different visualisations, build an interactive dashboard, use maps, and present information through a Tableau story. The project also taught me the importance of checking data quality and verifying findings before drawing conclusions.

10. Tools and Technologies Used

- Tableau Desktop
- Tableau Public
- Global Fuel Prices Dataset
- Spreadsheet software (if used during data preparation)

11. Learning Outcomes

The project helped me learn how to organise data for analysis, choose suitable chart types, create maps and line charts, design an interactive dashboard, compare categories, and communicate findings clearly.

12. References

1. Global Fuel Prices Dataset:
https://drive.google.com/file/d/15zw6SL_494VYIkGYEvY68YIeF9Gw8rCG/view?usp=sharing